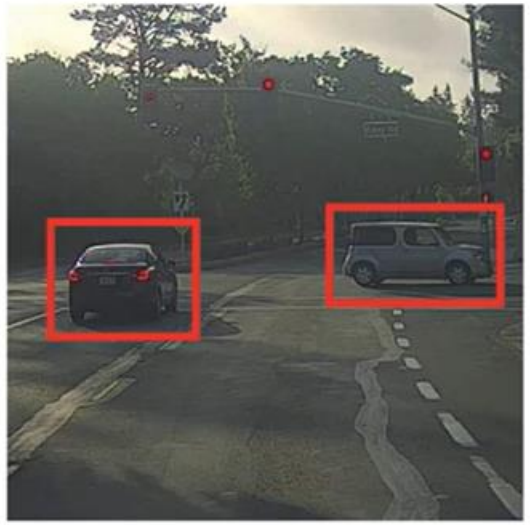
Lecture08
YOLO

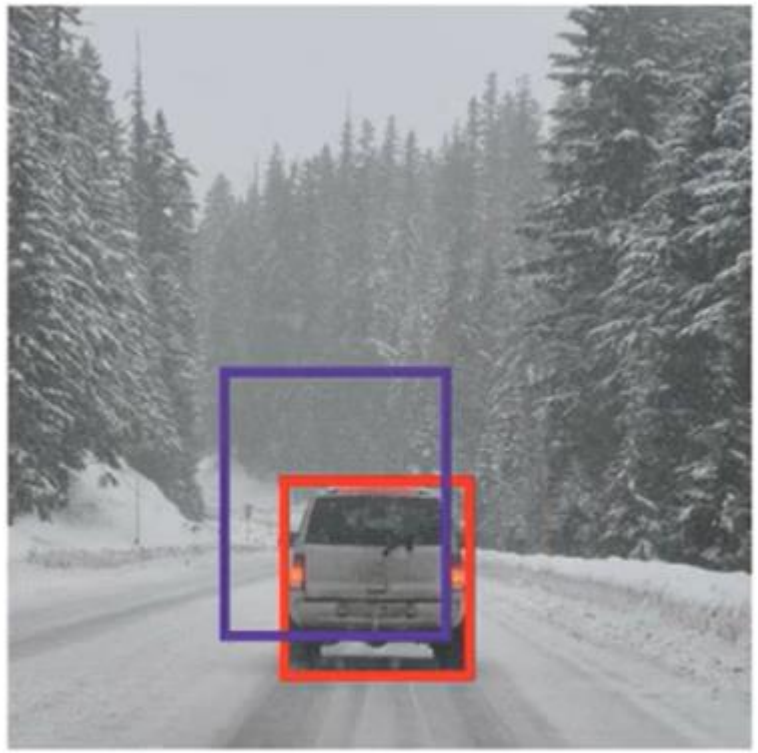
Sliding window detection



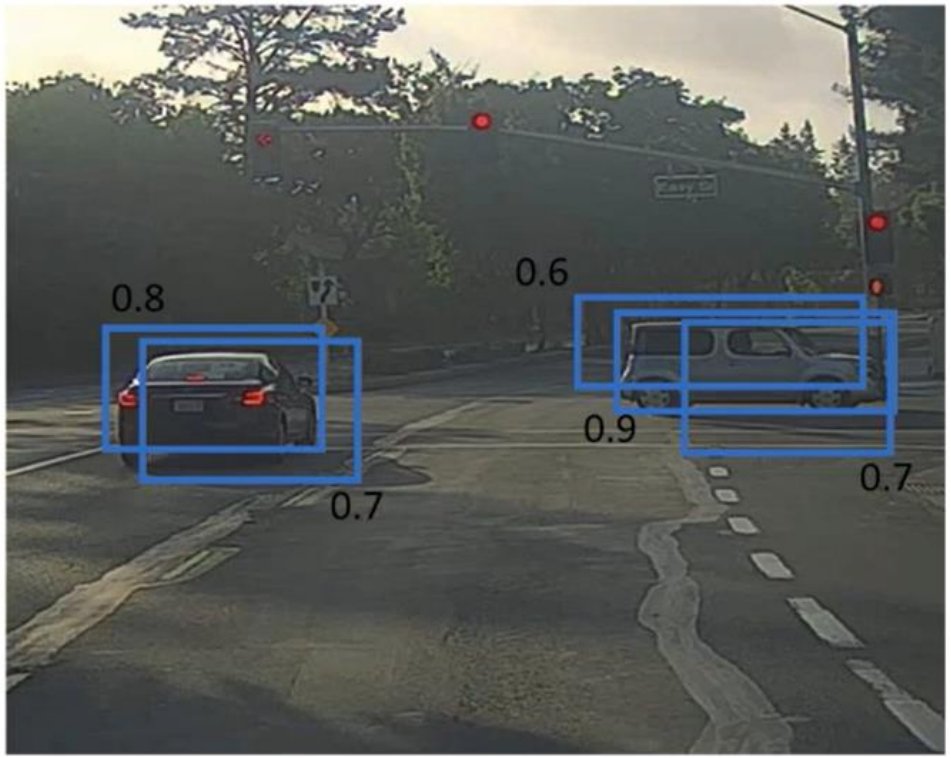
Training set:

x	y
	1
	1
	1
	0

Intersection over union (IoU)



Non-max supression



Objetos traslapados

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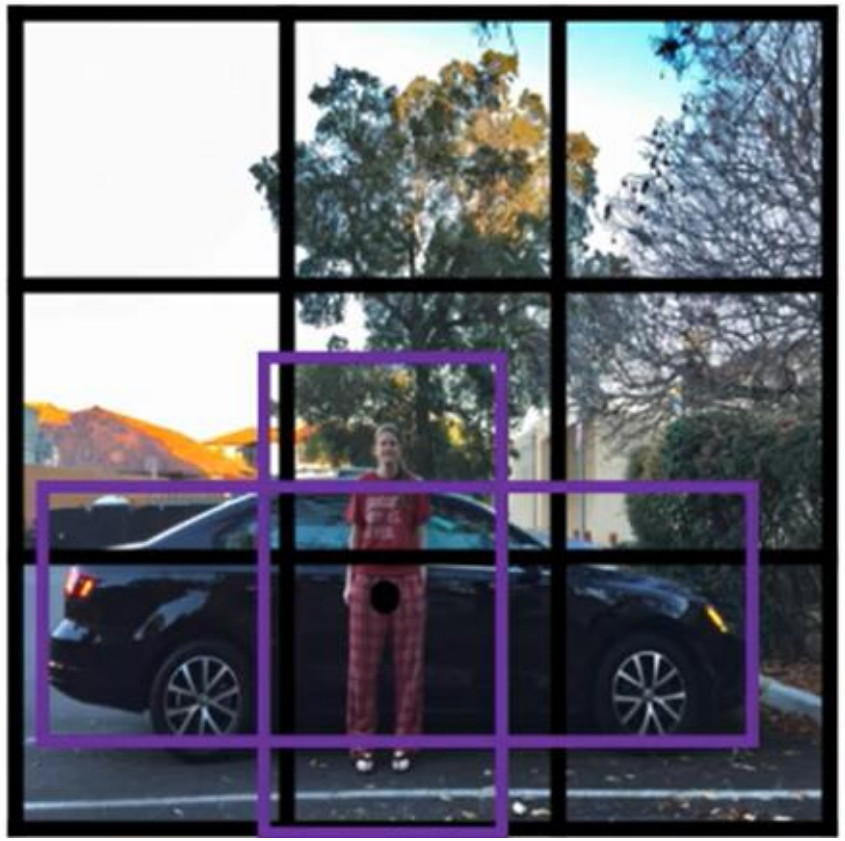
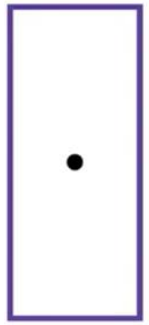


Anchor box 1:



Anchor box 2:

$$y = \begin{bmatrix} P_c^{(1)} \\ b_x^{(1)} \\ b_y^{(1)} \\ b_h^{(1)} \\ b_w^{(1)} \\ c_1^{(1)} \\ c_2^{(1)} \\ \vdots \\ c_j^{(1)} \\ P_c^{(2)} \\ b_x^{(2)} \\ b_y^{(2)} \\ b_h^{(2)} \\ b_w^{(2)} \\ c_1^{(2)} \\ c_2^{(2)} \\ \vdots \\ c_j^{(2)} \end{bmatrix}$$

[Redmon et al., 2015. You Only Look Once: Unified, Real-Time Object Detection]

Anchor box (caja ancla)

Algoritmo de caja ancla

Anteriormente: Cada objeto en la imagen de entrenamiento se asigna a la celda de la cuadrícula que contiene el punto medio de ese objeto.

Con dos cajas ancla: Cada objeto en la imagen de entrenamiento se asigna a la celda de la cuadrícula que contiene el punto medio del objeto y a la caja ancla para la celda de la cuadrícula con el mayor IoU (Intersección sobre Unión).

YOLO

1 - pedestrian

2 - car

3 - motorcycle

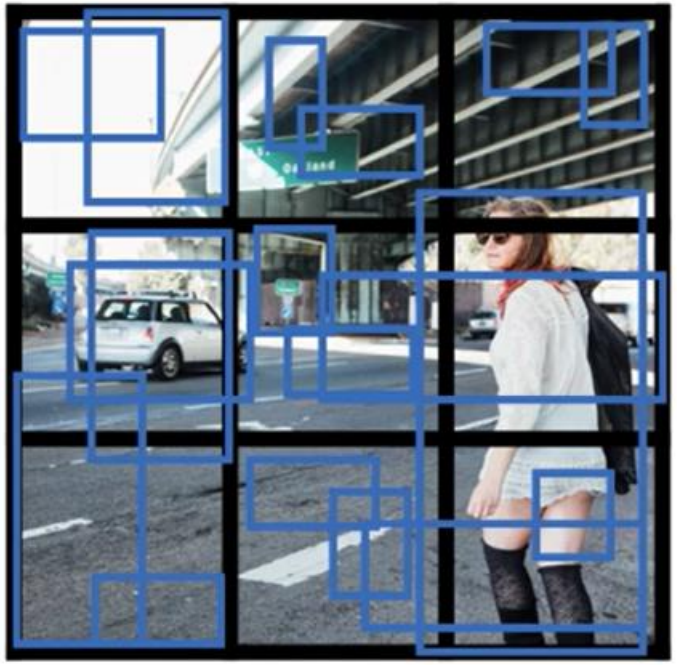
$$y = \begin{bmatrix} p_c \\ b_x \\ b_y \\ b_h \\ b_w \\ c_1 \\ c_2 \\ c_3 \\ p_c \\ b_x \\ b_y \\ b_h \\ b_w \\ c_1 \\ c_2 \\ c_3 \end{bmatrix} \quad \begin{bmatrix} 0 \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \\ 0 \\ ? \\ ? \\ ? \\ ? \end{bmatrix} \quad \begin{bmatrix} 0 \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \\ 1 \\ b_x \\ b_y \\ b_h \\ b_w \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbf{1}_{ij}^{\text{obj}} \left[(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2 \right]$$
$$+ \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbf{1}_{ij}^{\text{obj}} \left[\left(\sqrt{w_i} - \sqrt{\hat{w}_i} \right)^2 + \left(\sqrt{h_i} - \sqrt{\hat{h}_i} \right)^2 \right]$$
$$+ \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbf{1}_{ij}^{\text{obj}} (C_i - \hat{C}_i)^2$$
$$+ \lambda_{\text{noobj}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbf{1}_{ij}^{\text{noobj}} (C_i - \hat{C}_i)^2$$
$$+ \sum_{i=0}^{S^2} \mathbf{1}_{ij}^{\text{obj}} \sum_{c \in \text{Classes}} (p_i(c) - \hat{p}_i(c))^2$$

y is $3 \times 3 \times 2 \times 8$

[Redmon et al., 2015. You Only Look Once: Unified, Real-Time Object Detection]

Non-max supressed outputs



Non-max supression para cada clase